Quantity	Unit	Typical value (Ar)
length	σ	3.4×10^{-10} m
mass	m	6.69×10^{-26} Kg
energy	ϵ	1.65×10^{-21} J
time	$\sigma\sqrt{\mathrm{m}/\epsilon}$	2.17×10^{-12} s
velocity	$\sqrt{\epsilon/\mathrm{m}}$	1.57×10^{2} m/s
force	ϵ/σ	4.85×10^{-12} Newtons
temperature	$\epsilon/\mathrm{k_B}$	120K
pressure	ϵ/σ^3	4.20×10^{7} Newtons/m^2

Fig. 1.13 Table of units and their values for a model of argon (Ar)

then the equations become

$$\frac{d^2\boldsymbol{Q}_i}{d\tau^2} = -\sum_{\substack{j=1 \\ j\neq i}}^{\mathrm{N}} \frac{\hat{\varphi}'_{\mathrm{LJ}}(R_{ij})}{R_{ij}}(\boldsymbol{Q}_i - \boldsymbol{Q}_j).$$

This means that a natural choice for the unit of time is

$$\alpha^{-1} = \sigma\sqrt{\frac{m}{\epsilon}} \approx 2.17 \times 10^{-12}\mathrm{s}.$$

By using the units given here, we may work with a simplified form of the Lennard-Jones system involving unit masses and a parameter-independent potential energy function.

In Fig. 1.13 we summarize the typical units relevant for studies of argon. The role of temperature and pressure will be discussed in Chaps. 6–8. □

1.3 The Lagrangian and Generalized Coordinates

An N-body system is often formulated in terms of its *Lagrangian* $L = L(\boldsymbol{q}, \dot{\boldsymbol{q}})$: the difference between kinetic and potential energies. The somewhat mysterious Lagrangian function was introduced by Joseph Louis Lagrange. Its relevance becomes clear in formulating the *principle of least action*, a variational principle which gives an abstract formulation of the equations of motion. We defer introducing this derivation of the equations of motion until Chap. 2, where it is also used as a means of deriving a numerical method for molecular simulation.

For the system (1.3) with N atoms and $\mathrm{N_c} = 3\mathrm{N}$ configuration coordinates, the Lagrangian is

$$L \stackrel{\mathrm{def}}{=} \frac{\dot{\boldsymbol{q}}^T \boldsymbol{M} \dot{\boldsymbol{q}}}{2} - U(\boldsymbol{q}).$$

The equations of motion may be expressed in terms of the Lagrangian as:

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{\boldsymbol{q}}} = \frac{\partial L}{\partial \boldsymbol{q}}.$$

(Note that this must be interpreted in general as a set of $\mathrm{N_c} = 3\mathrm{N}$ equations, one for each atomic coordinate.) The real usefulness of the Lagrangian perspective comes when we introduce different types of coordinates to express the equations of motion; examples of this are given below.

When we introduce a smooth change of variables $\boldsymbol{q} = \boldsymbol{\Phi}(\boldsymbol{Q})$, where $\boldsymbol{\Phi} : \mathbb{R}^{\mathrm{N_c}} \to \mathbb{R}^{\mathrm{N_c}}$, this induces a corresponding transformation of the velocity vector by

$$\dot{\boldsymbol{q}} = \boldsymbol{\Phi}'(\boldsymbol{Q})\dot{\boldsymbol{Q}},$$

where $\boldsymbol{\Phi}'$ is the $\mathrm{N_c} \times \mathrm{N_c}$ Jacobian matrix of $\boldsymbol{\Phi}$. Then the Lagrangian of the system is transformed to

$$\tilde{L} = \frac{\dot{\boldsymbol{Q}}^T \boldsymbol{\Phi}'(\boldsymbol{Q})^T \boldsymbol{M} \boldsymbol{\Phi}'(\boldsymbol{Q})\dot{\boldsymbol{Q}}}{2} - U(\boldsymbol{\Phi}(\boldsymbol{Q})),$$

It is not essential for the change of variables to be a map from $\mathbb{R}^{\mathrm{N_c}}$ to $\mathbb{R}^{\mathrm{N_c}}$. For example, in a system with constraints, the change of variables might represent a parameterization of the manifold defined by the constraints. Then the mapping $\boldsymbol{\Phi}$ might be from $\mathbb{R}^{\mathrm{N_d}}$ to $\mathbb{R}^{\mathrm{N_c}}$, where $\mathrm{N_c} \geq \mathrm{N_d}$. An abstract set of configuration variables is referred to as a set of *generalized coordinates*. The introduction of generalized coordinates in an N-body system typically leads to a Lagrangian which can be viewed as quadratic in the velocities but with a "configuration-dependent mass matrix." We will assume that any such changes of variables are regular transformations in the sense that $\boldsymbol{\Phi}'$ is of full rank and the resulting generalized mass matrix is invertible.

Although generalized coordinates are often useful for understanding molecular systems, as we shall learn in later chapters, it is desirable to avoid the use of formulations with a configuration-dependent mass matrix since it complicates the design of numerical methods with good conservation properties. This is one reason that molecular simulation methods are typically described in a Cartesian coordinate framework.

1.4 Hamiltonian Formulation

In this section, we introduce the Hamiltonian description of the motion which is useful for discussing the underlying geometric structure associated to classical mechanics. We can understand the Hamiltonian formulation as a natural description of classical mechanics from several perspectives. First, it is based directly on the